GROUP 76 | AI101 MINI-PROJECT

Predicting Product Sales via Multi-Channel Advertising Spend

TV, Radio and Newspaper channels: a Machine Learning approach

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Repo: https://github.com/2026aib1092-hash/Mini-Project-Group-76.git

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2 Problem and Data

What and why

- Predict product sales from the advertising budget spent on TV, Radio and Newspaper.
- Helps decide how to split a marketing budget across channels.
- Shows which channel actually drives sales, so spend is not wasted.
- Task type: supervised regression (numeric target).

Dataset

- Kaggle: Advertising Sales Dataset (yasserh)
- Size: 200 rows, 4 numeric columns
- Budgets in thousands of dollars

Columns

TV	Radio	Newspaper	Sales
Input	Input	Input	Target

Split: 80% train / 20% test (random_state = 42)

3 Approach

DATA

- Load CSV (200 rows)
- Drop index column
- 0 missing values
- X: TV, Radio, Newspaper
- y: Sales
- 80/20 split (160 / 40)

MODEL

- Median imputer
- SelectKBest (k = 3)
- StandardScaler (fit on train)
- Multiple Linear Regression
- Trained on 160 rows

OUTPUT

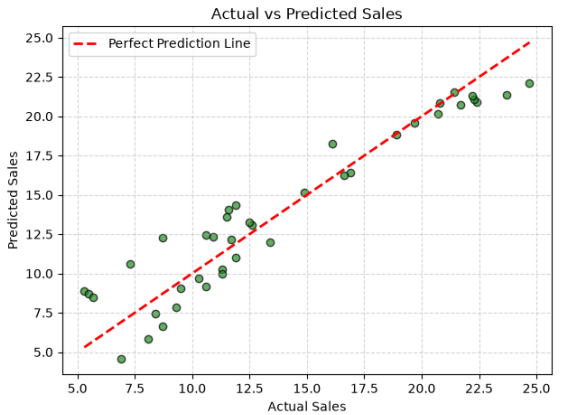
- Predicted sales
- R^2, MSE, RMSE, MAE
- Confusion matrix, precision, recall, F1 score
- Plots + demo prediction

Libraries used

pandas	NumPy	scikit-learn	matplotlib	os
load & clean data	arrays, median	model & metrics	scatter plots	CSV file path

sklearn modules: LinearRegression · train_test_split · SimpleImputer · SelectKBest · StandardScaler · metrics

4 Results



Confusion matrix

High vs low sales, median cut-off

	Pred. low	Pred. high
Actual low	**14** TN	**6** FP
Actual high	**1** FN	**19** TP

0.8994

R^2 score (test set)

Regression error

MSE	**3.1741**
RMSE	**1.7816**
MAE	**1.4608**

Demo prediction

```
Inference Prediction
(TV:200, Radio:30, Newspaper:15)
-> Estimated Sales: 17.6422
```

Predicted sales for TV 200, Radio 30, Newspaper 15

Classification metrics

0.8250 Accuracy	**0.7600** Precision
0.9500 Recall	**0.8444** F1 score

Learned coefficients

Intercept	**14.1000**
TV	**3.7642**
Radio	**2.7923**
Newspaper	**0.0560**

on standardised inputs

n = 40 test rows. High/Low split at the test median (positive = high). Run on the Kaggle Advertising Sales dataset.

5 Learnings

What worked

- $R^2 = 0.90$: the model explains ~90% of the variation in sales
- High/low accuracy 0.825, F1 0.84
- TV is the strongest driver (corr. 0.78), then Radio (0.58)

What didn't

- Newspaper adds very little (corr. 0.23, coef. 0.06)
- Precision 0.76: 6 of 25 high-sales calls were wrong
- Imputer and feature selection changed nothing: no NaNs, all 3 features kept

AI-use disclosure

AI tools were used for code assistance and to prepare these slides.

Slides were then edited by Vedansh.

Results were updated from the final notebook run.

Who did what

Vedansh Singla
PPT Lead: built the presentation and coordinated

Kallem Srivarsha
Video Lead: demo video

Jatin
Documentation Lead: README and report

Deepanshu Chahal
Execution Lead: ran the code and results